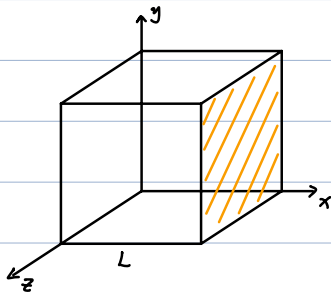


Kinetic theory of gases

Ideal gas

- Negligible molecular volume (compared to distance)
- Short-range force (active only during collision)
- Elastic collisions
- Identical molecules



Collision on the RHS wall: $v_x \rightarrow -v_x$

$$\Delta p_x = (-m v_x) - m v_x = -2m v_x$$

time interval between collisions:

$$\Delta t = \frac{2L}{v_x}$$

Force from a single molecule:

$$F_x = \frac{|\Delta p_x|}{\Delta t} = \frac{2m v_x}{2L/v_x} = \frac{m v_x^2}{L}$$

Pressure:

$$\begin{aligned} p &= \frac{F_{x1} + F_{x2} + \dots + F_{xN}}{L^2} \\ &= \frac{m}{L^3} (v_{x1}^2 + v_{x2}^2 + \dots + v_{xN}^2) \\ &= \frac{Nm}{L^3} \overline{v_x^2} \\ &= \frac{n N_A m}{L^3} \overline{v_x^2} \end{aligned}$$

$$\overline{v_x^2} \equiv \frac{1}{N} (v_{x1}^2 + v_{x2}^2 + \dots + v_{xN}^2)$$

$$n = \frac{N}{N_A}$$

$$\begin{cases} V = L^3 \\ v^2 = v_x^2 + v_y^2 + v_z^2 \end{cases}$$

$$\Rightarrow p = \frac{n N_A m}{3V} \overline{v^2}$$

$$\Rightarrow n R T = \frac{n N_A m}{3} \overline{v^2}$$

$$\begin{aligned} \Rightarrow v_{rms} &\equiv \sqrt{\overline{v^2}} = \sqrt{\frac{3RT}{N_A m}} \\ &= \sqrt{\frac{3k_B T}{m}} \end{aligned}$$

rms: root-mean-square (方均根)

Average translational kinetic energy per molecule:

$$K_{\text{avg}} = \frac{1}{2} m \overline{v^2} = \frac{1}{2} m v_{\text{rms}}^2$$

Since $v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$:

$$K_{\text{avg}} = \frac{3}{2} k_B T$$

- Temperature is a measure of the molecular kinetic energy !

Equipartition theorem (more serious proof later):

能量均分定理

Average kinetic energy per degree of freedom is $\begin{cases} \frac{1}{2} k_B T & \text{per molecule} \\ \frac{1}{2} R T & \text{per mole} \end{cases}$

e.g. 3 translational modes:

$$\frac{1}{2} m \overline{v_x^2} = \frac{1}{2} m \overline{v_y^2} = \frac{1}{2} m \overline{v_z^2} = \frac{1}{2} k_B T$$

Internal energy (monatomic gas)

$$U = \frac{3}{2} N k_B T$$

$$R = N_A k_B$$

$$= \frac{3}{2} n R T$$

Heat capacity (热容):

$$C \equiv \frac{\Delta Q}{\Delta T}$$

Specific heat capacity, abbr "specific heat" (比热、比热容):

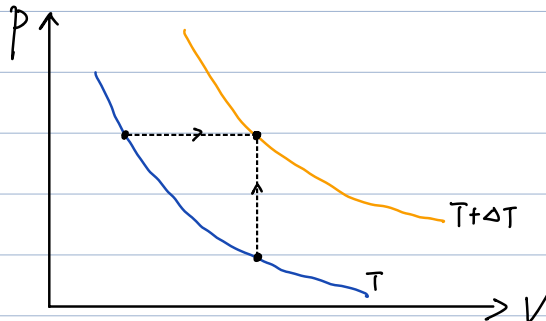
$$c \equiv \frac{C}{M}$$

(heat capacity per mass)

Molar heat capacity (摩尔热容):

$$c \equiv \frac{C}{n}$$

(heat capacity per mole)



Molar heat capacity at constant V (monoatomic gas):

$$c_v = \frac{C}{n} = \frac{1}{n} \left(\frac{\Delta Q}{\Delta T} \right)_v \quad \text{定容摩尔热容}$$

$$= \frac{1}{n} \left(\frac{dU}{dT} \right)_v$$

$$= \frac{3}{2} R$$

Molar heat capacity at constant p (monoatomic gas):

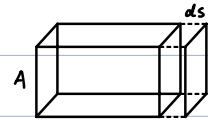
$$c_p = \frac{C}{n} = \frac{1}{n} \left(\frac{\Delta Q}{\Delta T} \right)_p \quad \text{定压摩尔热容}$$

$$= \frac{1}{n} \left(\frac{\Delta U + \Delta W}{\Delta T} \right)_p$$

$$\Delta W = F ds = p A ds = p dV$$

$$\Rightarrow c_p = \frac{1}{n} \left(\frac{dU}{dT} + p \frac{dV}{dT} \right)_p$$

$$\Rightarrow \boxed{c_p = c_v + R}$$



$$pV = nRT \Rightarrow p \left(\frac{dV}{dT} \right)_p = nR$$

Diatomic / Polyatomic gas

e.g. $\left\{ \begin{array}{l} \text{He : monoatomic} \\ \text{H}_2 : \text{diatomic} \\ \text{CO}_2 : \text{polyatomic} \end{array} \right.$

$$\left\{ \begin{array}{l} U = \frac{f}{2} N k_B T = \frac{f}{2} n R T \\ c_v = \frac{f}{2} R \\ c_p = c_v + R \end{array} \right. \quad \text{SAME at different } T$$

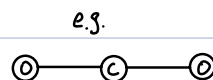
Degrees of freedom:

$$f = t + r + 2s$$

$\swarrow$ translation $\downarrow$ rotation $\searrow$ oscillation

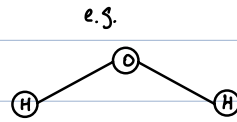
N -atom linear molecule:

$$\left\{ \begin{array}{l} t = 3 \\ r = 2 \\ s = 3N - 5 \end{array} \right.$$



N-atom nonlinear molecule:

$$\begin{cases} t = 3 \\ r = 3 \\ S = 3N - 6 \end{cases}$$



Q: Why 2S?

$$\text{oscillation} \begin{cases} \text{kinetic energy} & \frac{1}{2}RT \\ \text{potential energy} & \frac{1}{2}RT \end{cases}$$

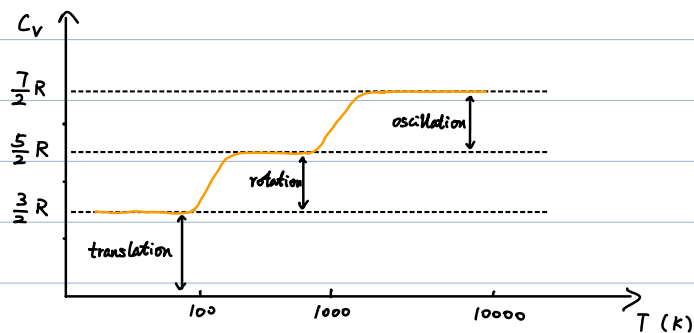
Example. H_2

$$t = 3, r = 2, S = 3N - 5 = 1$$

$$\Rightarrow f = t + r + 2S = 7$$

$$\Rightarrow C_v = \frac{7}{2}R$$

Reality:



(origin of discrepancy: quantum mechanics)

Statistics

Example.

Speeds of 10 particles: $v_1 = 0, v_2 = 1, \dots, v_{10} = 9 \text{ m/s}$

Average velocity: $\bar{v} = \frac{1}{N} \sum_i v_i = \frac{1}{10} (0 + 1 + \dots + 9) = 4.5 \text{ m/s}$

Root-mean-square velocity: $v_{\text{rms}} = \sqrt{\frac{1}{N} \sum_i v_i^2} = \sqrt{\frac{1}{10} (0^2 + 1^2 + \dots + 9^2)} \approx 5.34 \text{ m/s}$ ④

Probability distribution function $f(v)$:

Probability to find particle's velocity in $[v, v+dv]$ is $f(v)dv$.

Normalization:

$$\int_{-\infty}^{\infty} f(v) dv = 1$$

Average velocity:

$$\bar{v} = \int_{-\infty}^{\infty} v f(v) dv$$

Root-mean-square velocity:

$$v_{rms} = \sqrt{\int_{-\infty}^{\infty} v^2 f(v) dv}$$

Higher dimension:

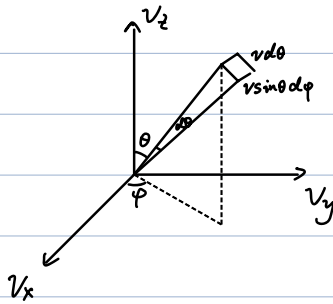
$$f(v)dv \rightarrow f(\vec{v}) dv_x dv_y dv_z$$

Maxwell distribution of velocity (速度):

Maxwell 1860

$$\begin{aligned} f(\vec{v}) &= f(v_x) f(v_y) f(v_z) \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}} \end{aligned}$$

$$v \equiv |\vec{v}|$$



Probability to find $\vec{v}$ in $[v, v+dv] \times [\theta, \theta+d\theta] \times [\varphi, \varphi+d\varphi]$:

$$f(\vec{v}) v^2 \sin \theta dv d\theta d\varphi$$

Isotropic (no dependence of $f(\vec{v})$ on θ, φ)

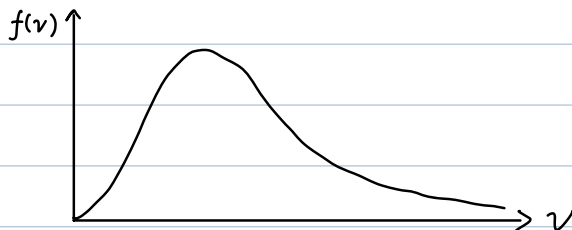
$\Rightarrow$ Probability to find $\vec{v}$ in $[v, v+dv]$: (any θ, φ)

$$\int_0^\pi d\theta \int_0^{2\pi} d\varphi f(\vec{v}) v^2 \sin \theta dv$$

$$= 4\pi f(\vec{v}) v^2 dv$$

Maxwell distribution of speed (速率):

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} e^{-\frac{mv^2}{2k_B T}} \cdot v^2$$



Characteristic speeds

Most probable speed v_p (最概然速率)

$$\frac{df(v)}{dv} = 0$$

$$\Rightarrow \frac{d}{dv} \left(e^{-\frac{mv^2}{2k_B T}} \cdot v^2 \right) = 0$$

$$\Rightarrow e^{-\frac{mv^2}{2k_B T}} \cdot \left(-\frac{mv}{k_B T} \right) v^2 + e^{-\frac{mv^2}{2k_B T}} \cdot 2v = 0$$

$$\Rightarrow \boxed{v_p = \sqrt{\frac{2k_B T}{m}}}$$

Average speed $\bar{v}$

$$\bar{v} = \int_0^{\infty} dv f(v) v$$

$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^{\infty} dv e^{-\frac{mv^2}{2k_B T}} \cdot v^3$$

$$\Rightarrow \boxed{\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}} \approx \sqrt{\frac{2.55 k_B T}{m}}$$

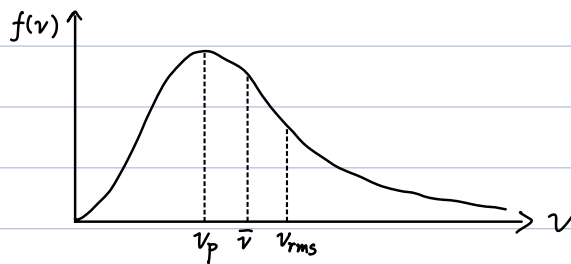
Root-mean-square speed v_{rms}

$$v_{rms}^2 = \int_0^{\infty} dv f(v) v^2$$

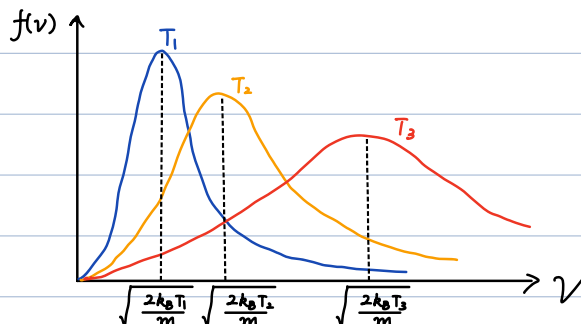
$$= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^{\infty} dv e^{-\frac{mv^2}{2k_B T}} \cdot v^4$$

$$= \frac{3k_B T}{m}$$

$$\Rightarrow \boxed{v_{rms} = \sqrt{\frac{3k_B T}{m}}}$$



Varying temperature



$$T_1 < T_2 < T_3$$

Maxwell-Boltzmann distribution (abbr "Boltzmann distribution")

$$f_{MB}(\vec{r}, \vec{v}) = f(x, y, z) f(v_x, v_y, v_z)$$

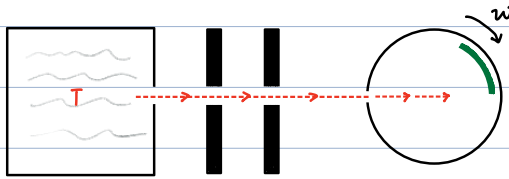
$$= \frac{1}{\int e^{-\epsilon_p(x,y,z)/k_B T} dx dy dz} e^{-\epsilon_p(x,y,z)/k_B T} \cdot \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-\frac{mv^2}{2k_B T}}$$

$$\propto \boxed{e^{-[\frac{1}{2}mv^2 + \epsilon_p(\vec{r})]/(k_B T)}}$$

$\epsilon_p(\vec{r})$: potential energy at $\vec{r}$.

Maxwell distribution - experimental verification

Otto Stern 1920



particles with different v hits
different part of the target.